WoSM Academy

MAT231-F27 · CALCULUS II

# Systems of linear differential equations

## Eigenvalues, phase portraits, and the matrix exponential

Dr. Ali Khudiyev
WoSM Academy

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# Last time

- $ay'' + by' + cy = 0$: the characteristic equation $ar^2 + br + c = 0$; real, double and complex roots; Euler's formula.
- Free and damped oscillations; forced oscillations and resonance.

Today: several unknown functions at once, $\mathbf{x}' = A\mathbf{x}$. The characteristic equation becomes the eigenvalue equation $\det(A - \lambda I) = 0$ — and Linear Algebra and Calculus meet.

# This lecture

01 Linear systems

02 Solutions from eigenvectors

03 Phase portraits and stability

04 The matrix exponential

05 Looking back

06 Summary and practice

Reading: Apostol, *Calculus, Volume 2*, Chapter 7.

01

# Linear systems

# Systems of differential equations

A **linear system with constant coefficients:** $\mathbf{x}'(t) = A\,\mathbf{x}(t)$, with $\mathbf{x}(t) \in \mathbb{R}^n$ and $A$ a constant $n \times n$ matrix.

**Example (Two rooms)**

Two rooms lose heat to the outside, and exchange heat with each other, at rate 1. With $x_1$, $x_2$ their temperatures above the outside temperature:

$$x_1' = -x_1 + (x_2 - x_1), \quad x_2' = -x_2 + (x_1 - x_2), \qquad A = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}.$$

# A second-order equation is a system

For $y'' + by' + cy = 0$, put $x_1 = y$ and $x_2 = y'$. Then $x_1' = x_2$ and $x_2' = -cx_1 - bx_2$:

$$\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -c & -b \end{pmatrix} \mathbf{x}, \qquad \det(A - \lambda I) = \lambda^2 + b\lambda + c,$$

the characteristic polynomial of Lecture 24.

- In the same way, any equation of order $n$ becomes a first-order system in $\mathbb{R}^n$.
- Numerical solvers (Euler's method and its refinements) work with first-order systems, so this is how higher-order equations are solved on a computer.

# Existence, uniqueness, superposition

**Theorem**

For every $\mathbf{x}_0 \in \mathbb{R}^n$, $\mathbf{x}' = A\mathbf{x}$ has exactly one solution with $\mathbf{x}(0) = \mathbf{x}_0$, and it is defined for all $t$.

- Sums and multiples of solutions are solutions, and $\mathbf{x} \mapsto \mathbf{x}(0)$ is a linear bijection onto $\mathbb{R}^n$: the solutions form an $n$-dimensional vector space.
- So we need $n$ independent solutions; every solution is a combination of them.
- Geometrically, solutions are the flow lines of the vector field $\mathbf{x} \mapsto A\mathbf{x}$ (Lecture 22). Their pictures in the $(x_1, x_2)$-plane form the **phase portrait**.

02

# Solutions from eigenvectors

# Straight-line solutions

Try $\mathbf{x}(t) = e^{\lambda t}\,\mathbf{v}$ with a constant vector $\mathbf{v} \neq \mathbf{0}$:

$$\mathbf{x}' = \lambda e^{\lambda t}\mathbf{v}, \qquad A\mathbf{x} = e^{\lambda t} A\mathbf{v}.$$

So $e^{\lambda t}\mathbf{v}$ is a solution exactly when $A\mathbf{v} = \lambda\mathbf{v}$: $\mathbf{v}$ is an **eigenvector** and $\lambda$ an **eigenvalue** of $A$ (Linear Algebra, Unit 17).

- The solution moves along the line through $\mathbf{v}$: outwards if $\lambda > 0$, inwards if $\lambda < 0$.
- The eigenvalues are the roots of $\det(A - \lambda I) = 0$, the analogue of the characteristic equation.

# The general solution

**Theorem**

If $A$ has $n$ linearly independent eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ with eigenvalues $\lambda_1, \dots, \lambda_n$, every solution of $\mathbf{x}' = A\mathbf{x}$ is

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t}\mathbf{v}_1 + \cdots + c_n e^{\lambda_n t}\mathbf{v}_n,$$

and the $c_i$ are found from $\mathbf{x}(0) = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$.

**Why**: with $V = (\mathbf{v}_1 \cdots \mathbf{v}_n)$, $A = V\Lambda V^{-1}$ (Linear Algebra, Unit 18). Put $\mathbf{x} = V\mathbf{y}$: then $\mathbf{y}' = \Lambda\mathbf{y}$, that is $y_i' = \lambda_i y_i$ for each $i$ — $n$ separate equations of Lecture 23, with $y_i = c_i e^{\lambda_i t}$.

Eigenvectors **decouple** the system. Each eigendirection evolves on its own.

# Example: two rooms

$A = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$: $\det(A - \lambda I) = (\lambda + 2)^2 - 1$, so $\lambda = -1$ with $\mathbf{v} = (1, 1)$, and $\lambda = -3$ with $\mathbf{v} = (1, -1)$.

Start with room 1 warm, room 2 not: $\mathbf{x}(0) = (1, 0) = \frac{1}{2}(1, 1) + \frac{1}{2}(1, -1)$. So

$$\mathbf{x}(t) = \frac{1}{2} e^{-t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2} e^{-3t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

- The **difference** $x_1 - x_2 = e^{-3t}$ disappears fast: the rooms equalise.
- The **average** $\frac{1}{2}(x_1 + x_2) = \frac{1}{2} e^{-t}$ decays slowly: both cool together.

# Complex eigenvalues

If $\lambda = \alpha + i\beta$ ($\beta \neq 0$) is an eigenvalue of a real matrix $A$, with eigenvector $\mathbf{v}$, the real and imaginary parts of $e^{\lambda t}\mathbf{v}$ are two real solutions (as in Lecture 24).

**Example.** $A = \begin{pmatrix} -1 & 2 \\ -2 & -1 \end{pmatrix}$: $\det(A - \lambda I) = (\lambda + 1)^2 + 4$, so $\lambda = -1 \pm 2i$. For $\lambda = -1 + 2i$, $\mathbf{v} = (1, i)$:

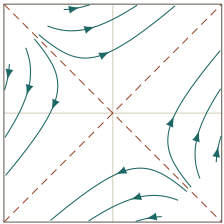
$$e^{\lambda t}\mathbf{v} = e^{-t}(\cos 2t + i \sin 2t)\begin{pmatrix} 1 \\ i \end{pmatrix} = e^{-t}\begin{pmatrix} \cos 2t \\ -\sin 2t \end{pmatrix} + i\, e^{-t}\begin{pmatrix} \sin 2t \\ \cos 2t \end{pmatrix}.$$

The solution with $\mathbf{x}(0) = (1, 0)$ is $e^{-t}(\cos 2t, -\sin 2t)$: it turns clockwise and shrinks — a **spiral.**

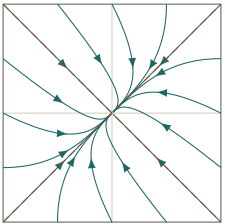
03

# Phase portraits and stability

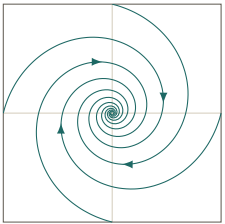
# Phase portraits



**Saddle:**
$\lambda = 3, -1.$
$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$



**Stable node:**
$\lambda = -1, -3.$
$A = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$



**Stable spiral:**
$\lambda = -\frac{1}{4} \pm i.$
$A = \begin{pmatrix} -1/4 & 1 \\ -1 & -1/4 \end{pmatrix}$

Dashed: the eigenvector lines, along which solutions move straight.

# Classification by trace and determinant

For a $2 \times 2$ matrix, $\lambda_1 + \lambda_2 = \operatorname{tr} A = T$ and $\lambda_1 \lambda_2 = \det A = D$, so $\lambda = \tfrac{1}{2}\left(T \pm \sqrt{T^2 - 4D}\right)$.

| Condition | Eigenvalues | Origin |
|---|---|---|
| $D < 0$ | real, opposite signs | saddle |
| $D > 0,\ T^2 \geq 4D$ | real, same sign | node |
| $D > 0,\ T^2 < 4D,\ T \neq 0$ | complex, $\operatorname{Re}\lambda = T/2$ | spiral |
| $D > 0,\ T = 0$ | purely imaginary | centre (closed orbits) |

Nodes and spirals are **stable** (attracting) if $T < 0$ and unstable if $T > 0$. Example: the damped spring $\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -k/m & -c/m \end{pmatrix}\mathbf{x}$ has $T = -c/m < 0$ and $D = k/m > 0$: stable, a spiral exactly when underdamped.

# Stability

**Theorem**

Every solution of $\mathbf{x}' = A\mathbf{x}$ tends to $\mathbf{0}$ as $t \to \infty$ if and only if every eigenvalue of $A$ has negative real part.

(For diagonalisable $A$: from $\sum_i c_i e^{\lambda_i t}\mathbf{v}_i$ and $|e^{\lambda t}| = e^{(\operatorname{Re}\lambda)\,t}$.)

**Linearisation (outlook)**

Near an equilibrium $\mathbf{x}^*$ of a nonlinear system $\mathbf{x}' = F(\mathbf{x})$, the Jacobian $DF(\mathbf{x}^*)$ (Lecture 12) usually decides stability.

Example: the gradient flow $\mathbf{x}' = -\nabla f(\mathbf{x})$ has Jacobian $-H_f$. At a strict local minimum $H_f$ is positive definite (Lecture 14), so the minimum attracts. Gradient descent is Euler's method for this flow.

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# The matrix exponential

# The matrix exponential

**Definition**

For a square matrix $A$: $\quad e^{tA} = I + tA + \dfrac{t^2A^2}{2!} + \dfrac{t^3A^3}{3!} + \cdots$

- The series converges for every $A$ and $t$: each entry of $t^kA^k/k!$ is at most $(|t|\,nM)^k/k!$ in absolute value, where $M$ is the largest $|a_{ij}|$ — compare with the series of $e^{|t|nM}$ (Lectures 2 and 5).
- Differentiating term by term (Lecture 5): $\frac{d}{dt}e^{tA} = Ae^{tA}$. So

$$\mathbf{x}(t) = e^{tA}\mathbf{x}_0$$

solves $\mathbf{x}' = A\mathbf{x}$, $\mathbf{x}(0) = \mathbf{x}_0$ — exactly like $x(t) = e^{at}x_0$ in one variable.

# Computing $e^{tA}$

- **Diagonalisable:** $A = V\Lambda V^{-1}$ gives $A^k = V\Lambda^k V^{-1}$, so

$$e^{tA} = V\ \mathrm{diag}\left(e^{\lambda_1 t}, \ldots, e^{\lambda_n t}\right) V^{-1}.$$

  Then $e^{tA}\mathbf{x}_0$ is the eigenvector solution from before.

- **A rotation:** $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ has $A^2 = -I$. Separating even and odd powers, as for Euler's formula:

$$e^{tA} = I\cos t + A\sin t = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}.$$

# When $A$ is not diagonalisable

$A = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} = \lambda I + N$, with $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $N^2 = 0$, has only one eigenvector direction.

Since $\lambda I$ and $N$ commute, $e^{t(\lambda I + N)} = e^{\lambda t}\, e^{tN}$, and the series for $e^{tN}$ stops after two terms:

$$e^{tA} = e^{\lambda t}\,(I + tN) = e^{\lambda t} \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

The factor $te^{\lambda t}$ is the double root of Lecture 24 again. In general every matrix has a Jordan form (Linear Algebra, Unit 24), and $e^{tA}$ contains terms $t^k e^{\lambda t}$.

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# Looking back

# The module in one slide

- **Series** (Units 1–8): convergence tests, power and Taylor series, generating functions, Stirling's formula.
- **Differential calculus in several variables** (Units 9–17): gradients, the chain rule and Jacobians, the Hessian and Taylor's formula, optimisation with and without constraints, gradient descent, implicit and inverse functions.
- **Integration** (Units 18–22): double and triple integrals, change of variables, the Gaussian integral, line integrals.
- **Differential equations** (Units 23–25): first-order, second-order linear, and linear systems.

Some threads: power series gave $e^{i\theta}$ and $e^{tA}$; eigenvalues decided the Hessian test, the speed of gradient descent, Lagrange problems on the sphere, and stability today; the Jacobian determinant governed the chain rule, inverse functions and the change of variables.

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# Summary and practice

# Summary

- $\mathbf{x}' = A\mathbf{x}$: unique solutions for all $t$; an $n$-dimensional space of solutions.
- Eigenpairs give solutions $e^{\lambda t}\mathbf{v}$; with $n$ independent eigenvectors, $\mathbf{x} = \sum c_i e^{\lambda_i t}\mathbf{v}_i$. Complex eigenvalues: take real and imaginary parts.
- $2 \times 2$ phase portraits: saddle, node, spiral, centre; classified by $\operatorname{tr} A$ and $\det A$. Stable iff all eigenvalues have negative real part.
- The matrix exponential: $\mathbf{x}(t) = e^{tA}\mathbf{x}_0$, with $e^{tA} = Ve^{t\Lambda}V^{-1}$ when $A$ is diagonalisable.

# Practice

1. Solve $\mathbf{x}' = A\mathbf{x}$ with $A = \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix}$ and $\mathbf{x}(0) = (3, 2)$.
2. Classify the origin for $\begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$, $\begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix}$, $\begin{pmatrix} 1 & -5 \\ 1 & -1 \end{pmatrix}$.
3. Solve $x' = y$, $y' = -4x$ with $x(0) = 1$, $y(0) = 0$. What curve does $(x(t), y(t))$ trace?
4. Write $y'' + 3y' + 2y = 0$ as a system, and compare the eigenvalues of its matrix with the roots of the characteristic equation.
5. Compute $e^{tA}$ for $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, and solve $\mathbf{x}' = A\mathbf{x}$.
6. For which $k$ do all solutions of $\mathbf{x}' = \begin{pmatrix} -1 & k \\ 1 & -1 \end{pmatrix}\mathbf{x}$ tend to $\mathbf{0}$?
7. Show directly that if $A\mathbf{v} = \lambda\mathbf{v}$, then $e^{\lambda t}\mathbf{v}$ solves $\mathbf{x}' = A\mathbf{x}$.

# Answers

1. $\lambda = 2$, $\mathbf{v} = (1,1)$; $\lambda = 3$, $\mathbf{v} = (2,1)$; $(3,2) = (1,1) + (2,1)$: $\mathbf{x} = e^{2t}(1,1) + e^{3t}(2,1)$.
2. $D = -4$: saddle. $T = -6$, $D = 8$, $T^2 > 4D$: stable node. $T = 0$, $D = 4$: centre.
3. $x = \cos 2t$, $y = -2\sin 2t$: the ellipse $x^2 + \frac{y^2}{4} = 1$, traversed clockwise.
4. $\begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}$: $\lambda^2 + 3\lambda + 2 = 0$, so $\lambda = -1, -2$ — the same as the roots of $r^2 + 3r + 2 = 0$.
5. $A^2 = 0$: $e^{tA} = I + tA = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$; $x_1 = x_1(0) + t\,x_2(0)$, $x_2 = x_2(0)$.
6. $T = -2 < 0$; need $D = 1 - k > 0$: $k < 1$.
7. $(e^{\lambda t}\mathbf{v})' = \lambda e^{\lambda t}\mathbf{v} = e^{\lambda t}A\mathbf{v} = A(e^{\lambda t}\mathbf{v})$.

# The end of the module

This was the last unit of Calculus II.

- **Final exam**: written, 3 hours, at the end of term (45% of the module). It tests computation (series, derivatives and integrals in several variables), the choice of method, and short proofs about convergence.
- **What uses this module**: multiple integrals and the Gaussian integral in Probability Theory and Statistics (joint densities); series and generating functions in the analysis of algorithms; gradients, Hessians and gradient descent in optimisation and machine learning.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.